A Diagrammatic Proof of the Haldane Gap via Spacetime Tensor Networks

(Dated: October 8, 2026)

Following the recent machine-assisted resolution of the Haldane conjecture for periodic and boundary-field spin-1 Heisenberg chains by OpenAI [1, 2], we show that the analytical derivations can be recast as elementary tensor-network contractions. Formulating the system as an exact continuous-time thermal matrix product operator exposes the underlying spacetime geometry. The step from a finite-scale estimate to a uniform gap then reduces to a contractive scale-doubling recurrence on planar diagrams, under three pages long and formalized in Lean 4 with the two numerical inputs as its only unproved hypotheses. The finite-scale estimate itself is taken unchanged from Ref. [1], whose certificate uses spectral filters and character projections. This illustrates how diagrammatic representations can clarify and compress machine-generated proofs in quantum many-body physics.

We consider the antiferromagnetic spin-1 Heisenberg chain with periodic boundary conditions $\boldsymbol{S}_{L+1} = \boldsymbol{S}_1$ and $J = k_B = 1$:

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}. \tag{1}$$

All system sizes are even and at least four. Let $E_0(L) \le E_1(L) \le \cdots$ include multiplicities and define the gap $\Delta_L = E_1(L) - E_0(L)$, the step from the ground level to the first excited level, . For the Gibbs weights $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$,

$$\frac{w_1}{w_0} = e^{-\beta\Delta_L} \le e^{-c\beta} \quad \Longrightarrow \quad \Delta_L \ge c. \tag{2}$$

The ratio decays in β like  ; the task is to make its rate c independent of L:

Theorem 1 (Periodic gap [1])**.** *For every even $L \ge 2304$, the chain (1) has a unique ground state and $\Delta_L > (\log 20)/784 \simeq 0.003821$.*

The spectral gap of the chain (1) was established by OpenAI [1] using machine-assisted proof search over a thirty-page analytical construction. That proof relies on coordinate expansions, degree-12 polynomial spectral filters, and character decompositions across dihedral symmetry sectors. Automated theorem provers naturally explore such symbolic coordinate representations, which can accumulate intermediate algebraic machinery over long deduction horizons.

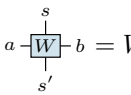
Here, we show that the underlying mechanism is fundamentally geometric. Representing the partition function as a continuous-time spacetime tensor network makes the dual spatial transfer operator, reflection Hermiticity, and scale-doubling contraction immediately manifest. The derivation proceeds in three stages. First, an exact continuous-time thermal MPO yields a self-adjoint spatial transfer operator. Second, spacetime contraction identities couple spatial and thermal purities, while scale doubling quadratically suppresses purity defects under joint iteration; we call this self-improving iteration the *spacetime bootstrap*. Third, ℓ^p moment monotonicity extends the bound from dyadic scales to all intermediate even lengths to establish Eq. (2).

Every step below is a contraction of one tensor network: horizontal legs — are virtual (space), vertical double legs ‖ are physical (imaginary time), joined legs are contracted, closed loops are traces [3, 4], and a labeled box is a product of tensors. Blue marks thermal objects ρ and red marks spatial ones σ. Appendix A applies the same networks to an open chain with boundary fields, where the spacetime is a cylinder rather than a torus; it reuses every identity below and adds only boundary vectors.

Thermal MPO.— To obtain a real symmetric transfer operator, use the Cartesian spin basis $s, s' \in \{x, y, z\}$. The matrices $(S^\alpha)_{ss'} = -i\epsilon_{\alpha ss'}$ give real operators $G_\alpha = iS^\alpha$, $\|G_\alpha\| = 1$, and $-H_L = \sum_{j,\alpha} G_{\alpha,j} G_{\alpha,j+1}$. The variable τ below is imaginary time. Because the ordered region $0 < \tau_1 < \cdots < \tau_k < \beta$ has volume $\beta^k/k!$, the exponential series of any matrix A can be written as a sum over ordered event times,

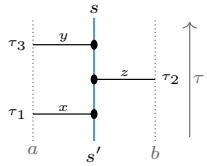
$$e^{\beta A} = \sum_{k\ge 0} \int_{0<\tau_1<\cdots<\tau_k<\beta} d\tau_1 \cdots d\tau_k \, \underbrace{A \cdots A}_{k}. \tag{3}$$

For $A = -H_L$, each factor selects one bond and one label α, so each term of Eq. (3) is a set of *bond events* (τ, α), drawn as rungs between the worldlines of neighboring sites. Each bond event (τ, α) in the exponential expansion inserts G_α at both endpoints. Let a, b record these events on the two bonds adjacent to a site, with $0 < \tau < \beta$. The local MPO tensor is



$$= W_\beta^{ss'}(a,b) = \left\langle s \middle| \mathcal{T}_\tau \prod_{(\tau,\alpha)\in a\cup b} G_\alpha \middle| s' \right\rangle, \tag{4}$$

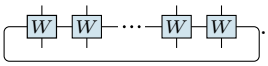
where $\mathcal{T}_\tau$ places later times to the left. Inside the box, W is the site's worldline read from bottom to top: for $a = \{(\tau_1, x), (\tau_3, y)\}$, $b = \{(\tau_2, z)\}$ and $\tau_1 < \tau_2 < \tau_3$, the tensor $W_\beta^{ss'}(a,b)$ is



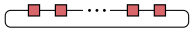
$$= \langle s|G_y G_z G_x|s'\rangle. \tag{5}$$

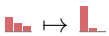
Operators on different sites commute, so once the events on every bond are fixed, the time-ordered product over

the ring splits into one such worldline per site. The events on a bond therefore play the role of the spin between two transfer matrices of the classical Ising chain, $Z_n = \operatorname{Tr} T^n$, with the 2×2 matrix T replaced by an operator on event histories. Summing event labels and integrating ordered times defines a measure μ_β of mass $e^{3\beta}$, with unit mass on the empty history. Contracting n tensors around the ring recovers the absolutely convergent exponential series as an exact MPO on the virtual space $L^2(\mu_\beta)$, without bond truncation [3, 4]:

$e^{-\beta H_n} =$  . (6)

Tracing the physical index,  gives the spatial transfer kernel $X_\beta(a,b) = \sum_s W^{ss}_\beta(a,b)$. Since a and b enter Eq. (4) only through their union, $X_\beta(a,b) = X_\beta(b,a)$. This left-right symmetry is what the classical Ising T has for free; for the quantum chain it holds because each event inserts the same real matrix at both ends of its bond, the special case of reflection positivity used here. It is real, symmetric, and bounded in absolute value by three, hence self-adjoint and Hilbert–Schmidt on $L^2(\mu_\beta)$. Closing the ring of transfer vertices gives, for even n,

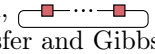
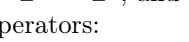
$Z_n(\beta) =$  $= \operatorname{Tr}_{\mathrm{v}} X^n_\beta = \sum_i |\lambda_i(\beta)|^n,$ (7)

where $\lambda_i(\beta)$ are the real eigenvalues of X_β, ordered by modulus, $|\lambda_0| \geq |\lambda_1| \geq \cdots$. Raising them to the nth power concentrates the weight on the largest:  for $n = 4$.

Purity and excited-state weight.— For a positive trace-one operator with eigenvalues $p_0 \geq p_1 \geq \cdots$, the purity obeys

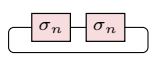
$$P = \sum_i p_i^2 \leq p_0 \sum_i p_i = p_0. \tag{8}$$

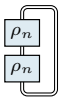
Thus $1-p_0 \leq 1-P$: a purity near one forces a spectrum like  with a single dominant bar. For the Gibbs state, $P > 1/2$ implies a unique ground state, with excited-state population $1-w_0 \leq 1-P$.

To propagate this bound, cut the ring of Eq. (7) at one bond,  $\mapsto$  , and normalize both the transfer and Gibbs operators:

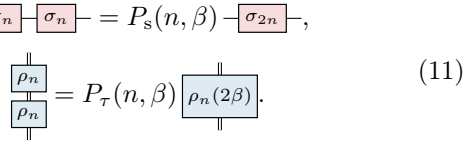
$$\sigma_n(\beta) = \frac{\text{---}^{n}\text{---}}{Z_n(\beta)} = \frac{X^n_\beta}{Z_n(\beta)}, \qquad \rho_n(\beta) = \frac{e^{-\beta H_n}}{Z_n(\beta)}. \tag{9}$$

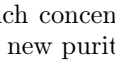
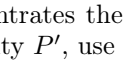
Both operators are positive with unit trace; denote their eigenvalues by p_i and w_m, respectively. Their purities join two copies, along space for σ_n and along imaginary time for ρ_n. Drawing $Z_n(\beta)$ as a spacetime torus  of width n and height β (dashed edges identified), both are one ratio of tori:

$P_{\mathrm{s}}(n,\beta) = \operatorname{Tr}_{\mathrm{v}} \sigma_n^2 =$  $= \dfrac{Z_{2n}(\beta)}{Z_n(\beta)^2} =$  ,

$P_\tau(n,\beta) = \operatorname{Tr} \rho_n^2 =$  $= \dfrac{Z_n(2\beta)}{Z_n(\beta)^2} =$  . (10)

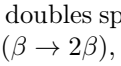
Squaring.— Squaring σ_n doubles the length; squaring ρ_n doubles the inverse temperature:

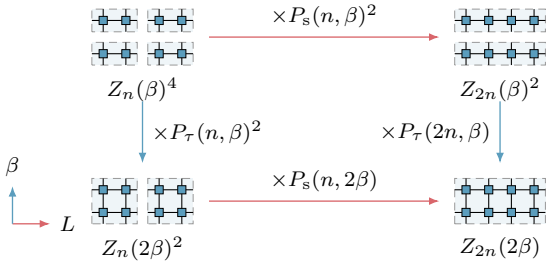
 (11)

Both replace the normalized eigenvalues p_i by p_i^2/P, which concentrates the spectrum:  $\mapsto$  . To bound the new purity P', use $1-P = 2\sum_{i<k} p_i p_k$:

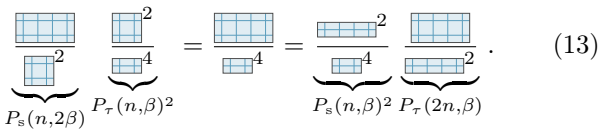
$$1-P' = \frac{2\sum_{i<k} p_i^2 p_k^2}{P^2} \leq \frac{(1-P)^2}{2P^2} = f(1-P), \qquad f(z) = \frac{z^2}{2(1-z)^2}. \tag{12}$$

Dropping cross terms from $(\sum_{i<k} p_i p_k)^2$ gives the inequality. The function f increases on $[0,1)$, with $f(z) \sim z^2/2$: squaring roughly squares the purity defect $1-P$, keeping it far below the diagonal,  for $0 \leq z \leq .3$.

Spacetime coupling identities.— While squaring independently doubles space ($n \to 2n$),  $\mapsto$  , or imaginary time ($\beta \to 2\beta$), $\mapsto$  , establishing a uniform gap requires doubling both variables simultaneously. Contracting four copies of the partition function $Z_n(\beta)$ in either spatial or temporal order yields the composite torus $Z_{2n}(2\beta)$. The networks are drawn to scale; red arrows join copies in space, blue arrows in time, and each arrow carries the purity factor it multiplies by:



Writing each purity as its ratio of tori, Eq. (10), the two paths cancel to the same quotient:

 (13)

Solving for one factor, and repeating at length $2n$, gives

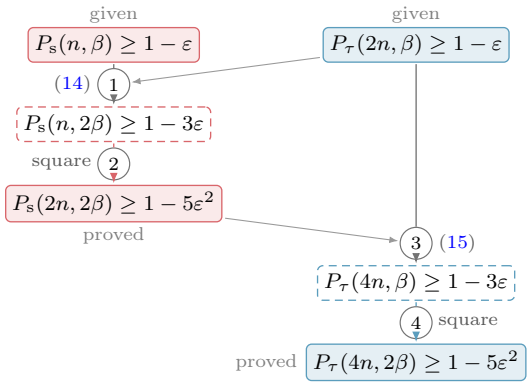
$$P_{\mathrm{s}}(n,2\beta) = \frac{P_{\mathrm{s}}(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2}, \tag{14}$$

$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_{\mathrm{s}}(2n,2\beta)}{P_{\mathrm{s}}(2n,\beta)^2}. \tag{15}$$

Doubling estimate.— Each application doubles the torus in both directions, ▫▭▭, and roughly squares the purity defect. For even n and $0 \le \varepsilon \le 1/100$,

$$\begin{aligned} & P_{\mathrm{s}}(n,\beta),\ P_\tau(2n,\beta) \ge 1-\varepsilon \\ \Longrightarrow\quad & P_{\mathrm{s}}(2n,2\beta),\ P_\tau(4n,2\beta) \ge 1-5\varepsilon^2. \end{aligned} \tag{16}$$

The scale-doubling update (16) is established by four moves, alternating between spacetime coupling and operator squaring. Read the diagram below from top to bottom: each box is one bound, red for P_{s} and blue for P_τ; solid boxes are given or proved, dashed boxes intermediate. Each move is a numbered circle labeled by the identity it uses; the grey arrow entering a circle is the second bound that move needs:



The proved pair is the given pair with $(n,\beta) \to (2n,2\beta)$, so it is the given pair of the next scale, which is why the estimate can be iterated.

Move 1 (coupling). Since $P_\tau(n,\beta) \le 1$, Eq. (14) gives $P_{\mathrm{s}}(n,2\beta) \ge (1-\varepsilon)^3 \ge 1-3\varepsilon$. *Move 2* (squaring in space). For even n, $\sigma_n(2\beta)$ has nonnegative spectrum and squares to $\sigma_{2n}(2\beta)$, so Eq. (12) and monotonicity of f give $P_{\mathrm{s}}(2n,2\beta) \ge 1-\delta$, where

$$\delta = f\big(1-(1-\varepsilon)^3\big) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3}-1\big]^2 \le 5\varepsilon^2 \le \varepsilon. \tag{17}$$

Here $1-(1-\varepsilon)^3 \le 3\varepsilon$ and $9/[2(1-3\varepsilon)^2] \le 5$ for $\varepsilon \le 1/100$. *Move 3* (coupling). Eq. (15) with $P_{\mathrm{s}}(2n,\beta) \le 1$ and $\delta \le \varepsilon$ gives $P_\tau(4n,\beta) \ge (1-\varepsilon)^2(1-\delta) \ge (1-\varepsilon)^3$. *Move 4* (squaring in time). $\rho_{4n}(\beta)$ squares to $\rho_{4n}(2\beta)$, so $P_\tau(4n,2\beta) \ge 1-\delta \ge 1-5\varepsilon^2$. This proves Eq. (16).

Finite input and iteration.— Proposition 5.5 of Ref. [1] provides the base-scale estimates:

$$P_{\mathrm{s}}(2304,784),\ P_\tau(4608,784) \ge 0.99. \tag{18}$$

The energy shift used there cancels from the purity ratios. After $j \ge 0$ doublings, set

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}. \tag{19}$$

Since $\varepsilon_0 = 1/100$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$, Eq. (16) gives $P_{\mathrm{s}}(n_j,\beta_j), P_\tau(2n_j,\beta_j) \ge 1-\varepsilon_j$ by induction on j: each application runs the four moves above once. The defect falls doubly exponentially in j, $\varepsilon_0 = 10^{-2}$ → $\varepsilon_4 \approx 3\times10^{-22}$ (log scale), and so exponentially in inverse temperature, $\varepsilon_j = \frac{1}{5}\exp[-\beta_j(\log 20)/784]$, as needed in Eq. (2).

These estimates hold at the doubled lengths. To cover every even $L \ge 2304$, choose j with $n_j \le L \le 2n_j$. Monotonicity of the ℓ^p norms in Eq. (7) gives

$$\begin{aligned} Z_L(\beta_j) &\le Z_{n_j}(\beta_j)^{L/n_j}, \\ Z_L(2\beta_j) &\ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \end{aligned} \tag{20}$$

Substitution into the thermal-purity ratio gives

$$\begin{aligned} P_\tau(L,\beta_j) &\ge \big[P_\tau(2n_j,\beta_j)P_{\mathrm{s}}(n_j,\beta_j)^2\big]^{L/(2n_j)} \\ &\ge (1-\varepsilon_j)^3 \ge 1-3\varepsilon_j > \tfrac{1}{2}. \end{aligned} \tag{21}$$

Here $L/(2n_j) \le 1$, and both purities in brackets are at least $1-\varepsilon_j$.

Recovering the gap.— Order the eigenvalues of $\rho_L(\beta_j)$ as $w_0 \ge w_1 \ge \cdots$. By Eqs. (8) and (21), $w_0 \ge 1-3\varepsilon_j > 1/2$: the Gibbs spectrum looks like ▮▁, with the ground state alone above one half. Two ground states would show two equal top bars, ▮▮, neither above one half; so the ground state is unique. The remaining weight bounds w_1, and hence

$$e^{-\beta_j \Delta_L} = \frac{w_1}{w_0} \le \frac{1-w_0}{w_0} \le \frac{3\varepsilon_j}{1-3\varepsilon_j} < 20^{-2^j}. \tag{22}$$

The last inequality uses $3\varepsilon_j/(1-3\varepsilon_j) \le (60/97)20^{-2^j}$. Taking logarithms and dividing by $\beta_j = 784\,2^j$ gives, for every even $L \ge 2304$,

$$\boxed{\Delta_L > \frac{\log 20}{784} \simeq 0.003821.} \tag{23}$$

This proves Theorem 1, conditional on Eq. (18); in particular $\liminf_{L\to\infty,\ L\,\mathrm{even}} \Delta_L \ge (\log 20)/784 > 0$. The scale-doubling recurrence and the gap deduction are formally verified in Lean 4 (`lean/CertifiedGaps/PurityCriterion.lean`).

DISCUSSION: REPRESENTATION IN MACHINE-ASSISTED PROOFS

The uniform spectral gap theorems for the periodic and boundary-field spin-1 Heisenberg chains, together with the certified finite-scale numerical inputs, originate from the machine-assisted proofs by OpenAI [1, 2].

The original thirty-page proofs isolate eigenvalues and bridge scales through extensive coordinate constructions, including degree-12 polynomial spectral filters and dihedral character projectors, alongside lemmas that remain unused in the final inequality. Automated reasoning systems typically operate via local symbolic rewrite rules

in a fixed basis. Without the geometric and topological representations native to many-body physics, long-horizon symbolic search tends to accumulate representational overhead to ensure soundness.

In this work, the authors proposed to GPT-6 to examine the argument through spacetime tensor networks, and interactively digested the resulting structure. Tensor network diagrams provide the geometric inductive bias that symbolic coordinate expansions lack. By placing imaginary time and spatial extent on equal footing on a Euclidean torus, the self-adjoint spatial transfer operator and contractive scale-doubling iteration become elementary planar identities. This indicates that diagrammatic tensor networks can serve not only as a calculational tool, but as an effective distillation framework for interpreting, simplifying, and extracting the essential physical mechanisms from machine-generated mathematical proofs. All mathematical arguments and formal proofs were verified and adjudicated by the authors.

[1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Proposition 5.5 and Appendices A–C.

[2] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, manuscript dated September 24, 2026, Secs. 2–6 and Appendices A and B.

[3] W. Tang, H.-H. Tu, and L. Wang, Continuous matrix product operator approach to finite temperature quantum states, Phys. Rev. Lett. **125**, 170604 (2020).

[4] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, Matrix product operator representations, New J. Phys. **12**, 025012 (2010).

[5] T. Kennedy, Exact diagonalisations of open spin-1 chains, J. Phys.: Condens. Matter **2**, 5737 (1990).

Appendix A: Open chain with boundary fields

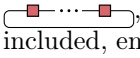
The networks above also bound the gap of the open chain with equal endpoint fields. On n spin-1 sites, with $\sum_\alpha (S^\alpha)^2 = 2I$, let

$$H_n^{\mathrm{s}} = \sum_{i=1}^{n-1} \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} - h(S_1^z + S_n^z), \qquad h = \frac{3}{5}. \tag{A1}$$

Theorem 2 (Boundary-field gap [2]). *For every odd $n \geq 1921$, the Hamiltonian (A1) has a unique ground state and $\Delta_n > (\log 10)/392 \simeq 0.005874$.*

Only one ingredient is new. On a ring, every virtual leg of Eq. (6) joins two tensors. On an open chain, the outer legs of the first and last sites have no partner, and a vector on the virtual space must be inserted there. This is the quantum counterpart of the open classical Ising chain, whose partition function is $u^{\mathsf{T}} T^{n-1} u$, with $u = (e^{\beta h}, e^{-\beta h})$ for a field h on each end spin.

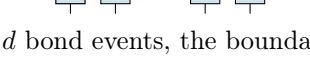
The fields are needed: without them, each end of the open spin-1 chain carries an effective spin $\frac{1}{2}$, and the four lowest states are split only by an amount exponentially small in n [5]. A field on an end spin enters the network as half a bond, whose events insert G_z on the end site and a number outside it [Eq. (A4) below].

The boundary vectors also change which eigenvectors of X_β the network sees. A periodic network is closed by a trace,  , in which every eigenvector, the leading one included, enters with weight one. An open chain is closed by two boundary vectors,  , which weight the leading eigenvector by its squared overlap with the boundary vector [Eq. (A8)]; this overlap vanishes if the two are orthogonal. The proof therefore tracks this *boundary overlap* alongside the thermal purity. As in Eqs. (2) and (8), it suffices to bound the equal-field thermal purity $P_\tau^{\mathrm{s}}(n,\beta) = \sum_m w_m^2$ from below, since

$$e^{-\beta\Delta_n} = \frac{w_1}{w_0} \leq \frac{1 - P_\tau^{\mathrm{s}}(n,\beta)}{P_\tau^{\mathrm{s}}(n,\beta)}. \tag{A2}$$

The scales of this appendix replace those of Eq. (19); they are fixed in Eq. (A20), and ε_j denotes the periodic purity defect of Eq. (B1).

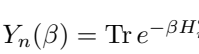
Boundary vectors.— The open chain uses the tensor W_β of Eq. (4), with the energy shift $\kappa = 700741/500000$ of Ref. [2] (called a there) absorbed as a factor $e^{-\kappa\beta}$, i.e. $H_n^\nu \to H_n^\nu + \kappa n I$; it cancels from every Gibbs state. Opening the ring of Eq. (6) and capping it with boundary vectors gives

$$e^{-\beta H_n^{\mathrm{s}}} = \text{  }. \tag{A3}$$

For a history a of d bond events, the boundary vector is

$$v_\beta(a) = \begin{cases} (ih)^d, & \text{all labels are } z, \\ 0, & \text{otherwise,} \end{cases} \qquad \|v_\beta\|^2 = e^{\beta h^2}, \tag{A4}$$

with value one on the empty history. Each event on a cap inserts a field on the end site: the left cap is conjugated, so its factor $(-ih)G_z = +hS^z$ adds $-hS_1^z$ to H, while the same vector on the right, $(ih)G_z = -hS^z$, adds $+hS_n^z$. The π rotation $P = \mathrm{diag}(1,-1,-1)$ about x flips every z label, so its virtual action U_P reverses the right field. The right cap alone therefore sets the field pattern of the trace:

$$Y_n(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{o}}} = \text{  }, \tag{A5}$$

$$Y_n^{\mathrm{s}}(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{s}}} = \text{}, \tag{A6}$$

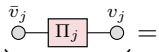
so $H_n^{\mathrm{o}} = H_n^{\mathrm{s}} + 2hS_n^z$; here X_β^n is the chain of Eq. (9), and Z_n of Eq. (7) is the periodic counterpart. The opposite-field chain has identical bra and ket caps, so its even moments are nonnegative. Simultaneous spin rotations of the histories act on $L^2(\mu_\beta)$ as real orthogonal operators commuting with X_β; in particular $[U_P, X_\beta] = 0$. Before the shift, the absolute traced terms sum to at most $3^n e^{\beta[3(n-1)+2h]}$, which justifies the regrouping.

Transfer operator spectrum.— At successive inverse temperatures $\beta_{j+1} = 2\beta_j$, the periodic estimates show that the leading eigenvalue of Eq. (7), $\lambda_j := \lambda_0(\beta_j)$, is simple and positive. Normalizing it to one:

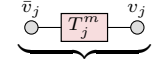
$$T_j := X_{\beta_j}/\lambda_j = \Pi_j + R_j, \qquad \Pi_j = |\psi_j\rangle\langle\psi_j|, \quad \|\psi_j\| = 1, \tag{A7}$$

where $\Pi_j R_j = R_j \Pi_j = 0$, so $T_j^m = \Pi_j + R_j^m$. Furthermore, $\|R_j\| < 1$, $U_P \Pi_j = \Pi_j$, and $0 < r_j := \lambda_{j+1}/\lambda_j^2 \leq 1$ (Appendix B).

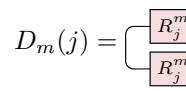
Boundary overlap and residual decomposition.— Put $v_j = v_{\beta_j}$. Inserting the leading projector disconnects the network:

$$\underbrace{\text{  }}_{c_j} = \text{} = |\langle\psi_j, v_j\rangle|^2. \tag{A8}$$

The overlap c_j is the squared projection of v_j onto ψ_j,  ; unlike a periodic trace, where $\mathrm{Tr}_{\mathrm{v}}\, \Pi_j = 1$, it vanishes when the two are orthogonal. For even $m \geq 2$,

$$\underbrace{\text{  }}_{B_m(j)} = \underbrace{\text{}}_{c_j} + \underbrace{\text{}}_{D_m(j)}. \tag{A9}$$

Folding establishes nonnegativity of the remainder:

$$D_m(j) = \text{  } = \|R_j^{m/2} v_j\|^2 \geq 0. \tag{A10}$$

Self-adjointness of R_j makes the lower row the mirror of the upper one, while $B_m = Y_m/\lambda_j^m > 0$. The boundary defect is defined by

$$e_m(j) := \frac{D_m(j)}{B_m(j)} = 1 - \frac{c_j}{B_m(j)}, \tag{A11}$$

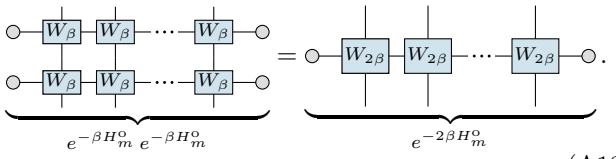
the grey share of the contraction,  , drawn at the starting value $e_{142}(2) \leq .14$ of Eq. (C1). Any $e_m < 1$ guarantees a nonzero overlap with the leading eigenvector.

Spatial remainder suppression.— For even $m \geq k$, absorbing $k/2$ residual factors into each boundary gives $w = R_j^{k/2} v_j$ and

$$D_m(j) = \bar{w} \text{—} R_j^{m-k} \text{—} w \leq \|R_j\|^{m-k} \underbrace{\|w\|^2}_{D_k(j)}. \tag{A12}$$

Spatial growth suppresses the remainder geometrically, by $\|R_j\|^{m-k}$, while leaving c_j invariant, ensuring $c_j \leq B_m(j) \leq B_k(j)$.

Thermal operator composition.— Composition in imaginary time, $e^{-\beta H_m^{\circ}} e^{-\beta H_m^{\circ}} = e^{-2\beta H_m^{\circ}}$, couples temperatures:



(A13)

This holds as an operator identity on the physical Hilbert space without requiring sitewise virtual fusion. Normalizing the trace defines the opposite-field thermal purity:

$$P_\tau^{\circ}(m, \beta_j) := \frac{Y_m(\beta_{j+1})}{Y_m(\beta_j)^2} = \operatorname{Tr} \rho_m^{\circ}(\beta_j)^2 \in (0, 1], \tag{A14}$$

with $\rho_m^{\circ}(\beta) = e^{-\beta H_m^{\circ}}/Y_m(\beta)$. Factoring out leading eigenvalues links adjacent temperatures:

$$B_m(j+1) = P_\tau^{\circ}(m, \beta_j) r_j^{-m} B_m(j)^2. \tag{A15}$$

Coupled overlap recurrence.— Assuming $e_k(j) < 1$ at scale j, let primes denote $j+1$, choose even $m > k \geq 2$, and let $\|R_{j+1}\|^{m-k} \leq q < 1$. Spatial suppression at $j+1$ gives $D'_m \leq q D'_k$. Since c' is length-independent, $D'_k - D'_m = B'_k - B'_m$, implying

$$(1-q) D'_m \leq q(B'_k - B'_m), \qquad e'_m \leq \frac{q}{1-q} \left(\frac{B'_k}{B'_m} - 1 \right). \tag{A16}$$

Composition relates the ratio to the previous temperature:

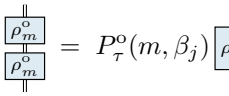
$$\begin{aligned} \frac{B'_k}{B'_m} &= \frac{P_\tau^{\circ}(k, \beta_j)}{P_\tau^{\circ}(m, \beta_j)} \left[\frac{B_k(j)}{B_m(j)} \right]^2 r_j^{m-k} \\ &\leq \frac{1}{P_\tau^{\circ}(m, \beta_j)[1 - e_k(j)]^2}, \end{aligned} \tag{A17}$$

using $P_\tau^{\circ}(k, \beta_j), r_j \leq 1$ and $B_m(j) \geq c_j = [1 - e_k(j)] B_k(j) > 0$. Because only the previous overlap appears in the denominator, $e'_m < 1$ establishes that the updated overlap remains positive. When $e_k(j)$ and $1 - P_\tau^{\circ}(m, \beta_j)$ are $\mathcal{O}(x)$, choosing $m - k$ with $q \leq \mathcal{O}(x)$ ensures that e'_m contracts quadratically as $\mathcal{O}(x^2)$.

Thermal purity propagation.— The boundary bound enables physical length doubling. For even M, Eq. (A15) yields

$$\frac{P_\tau^{\circ}(2M, \beta_j)}{P_\tau^{\circ}(M, \beta_j)} = r_j^M \underbrace{\frac{B_{2M}(j+1)}{B_M(j+1)}}_{\geq 1 - e_M(j+1)} \underbrace{\left[\frac{B_M(j)}{B_{2M}(j)} \right]^2}_{\geq 1}. \tag{A18}$$

The first bound retains c_{j+1}, while the second uses Eq. (A12). Because $r_j^M \approx 1$ [Eq. (B2)], purity remains controlled upon doubling length. Squaring the Gibbs operator advances temperature, as in Eq. (11):



(A19)

so the squaring bound (12) at $m = 2M$ drives $1 - P_\tau^{\circ}$ to $\mathcal{O}(x^2)$.

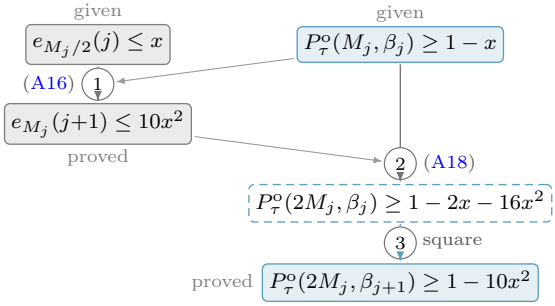
Scale induction.— Coupling the overlap and purity recurrences enables simultaneous induction over physical length and inverse temperature. Using the scales of this appendix,

$$\beta_j = 49\, 2^{j-2}, \quad n_j = 120\, 2^{j-2}, \quad M_j = 2n_j \quad (j \geq 2), \tag{A20}$$

with $k_2 = 142$, $k_{j+1} = M_j$, and with periodic error ε_j bounded in Eq. (B1), induction establishes

$$\begin{aligned} &\varepsilon_j,\ e_{k_j}(j),\ 1 - P_\tau^{\circ}(M_j, \beta_j) \leq x_j \quad (j \geq 5), \\ &x_5 = .01, \qquad x_{j+1} = 10 x_j^2, \qquad x_j = 10^{-1-2^{j-5}}. \end{aligned} \tag{A21}$$

For $j \geq 3$, $k_j = M_j/2$, leaving $M_j/2$ sites to suppress remainders. One step $j \to j+1$ is three moves. In the diagram below, read from top to bottom, each box is one bound: grey for the boundary defect $e_k(j)$, blue for the opposite-field thermal purity P_τ°. Solid boxes are given at scale j or proved at scale $j+1$, the dashed box is intermediate, and the grey arrow entering a move is the second bound it uses; the bounds shown hold for $j \geq 5$ (Appendix C):



Move 1 is the overlap recurrence (A16)–(A17), move 2 the length doubling (A18), and move 3 the squaring (A19). Since $M_{j+1} = 2M_j$, the proved pair is the given pair of scale $j+1$, so the step repeats. The defect falls doubly exponentially, $x_5 = 10^{-2} \to x_9 = 10^{-17}$ (log scale); since $\beta_j = 392\, 2^{j-5}$, it decays exponentially in inverse temperature.

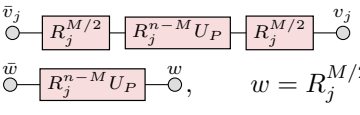
Equal boundary fields and odd lengths.— The opposite-field estimates extend to the equal-field Hamiltonian (A1) via the spin-rotation symmetry U_P. Next to the leading projector, the rotation operator acts trivially:

$$\text{—} \Pi_j \text{—} U_P \text{—} = \text{—} \Pi_j \text{—}. \tag{A22}$$

Thus $Q_j(n) := Y_n^{\mathrm{s}}(\beta_j)/\lambda_j^n$ shares the leading value c_j. For any even $M \leq n$, commuting U_P through the right-

hand tensors ($[U_P, R_j] = 0$) isolates the remainder:

$$Q_j(n) - c_j = \quad = \quad , \qquad w = R_j^{M/2} v_j. \tag{A23}$$



Because $\|R_j^{n-M} U_P\| \le 1$,

$$|Q_j(n) - c_j| \le \|w\|^2 = D_M(j), \tag{A24}$$

bounding the odd-length remainder by $D_M(j)$ without requiring positivity of R_j^n.

For $M_j \le n \le 2M_j$ and $M = M_j = k_{j+1}$, this yields

$$Q_j(n) \le B_M(j), \quad Q_{j+1}(n) \ge (1-2x_{j+1}) B_M(j+1) > 0. \tag{A25}$$

Dividing these contractions via Eq. (A15) gives the equal-field purity:

$$\begin{aligned} P_\tau^{\mathrm{s}}(n, \beta_j) = \frac{Y_n^{\mathrm{s}}(\beta_{j+1})}{Y_n^{\mathrm{s}}(\beta_j)^2} &\ge (1-2x_{j+1}) P_\tau^{\mathrm{o}}(M, \beta_j) r_j^{n-M} \\ &\ge (1-2x_{j+1})(1-x_j)(1-\varepsilon_j)(1-\varepsilon_{j+1}) \\ &\ge 1 - 2x_j - 26x_j^2 \ge 1 - 4x_j, \end{aligned} \tag{A26}$$

using $r_j^{n-M} \ge r_j^M$ [Eq. (B2)] and Appendix C bounds ($\varepsilon_j \le x_j$, $\varepsilon_{j+1} \le 6x_j^2$, $x_{j+1} = 10x_j^2$).

Proof of Theorem 2. The intervals $[M_j, 2M_j]$ cover every $n \ge M_5 = 1920$. For odd $n \ge 1921$, choose such a j. The largest Gibbs weight satisfies $w_0 \ge P_\tau^{\mathrm{s}}(n, \beta_j) \ge 1 - 4x_j > 1/2$: the spectrum looks like ⌊, so the ground state is unique. Equation (A2) gives

$$e^{-\beta_j \Delta_n} \le \frac{4x_j}{1-4x_j} \le \frac{5}{12}\, 10^{-2^{j-5}} < 10^{-2^{j-5}}. \tag{A27}$$

Taking logarithms and using $\beta_j = 392\, 2^{j-5}$ yields $\Delta_n > (\log 10)/392$. □

Appendix B: Transfer bounds at the open-chain scales

For the scales in Eq. (A20), Appendix A uses

$$\|R_j\|^{n_j} \le d(\varepsilon_j), \qquad d(\varepsilon) = \frac{\varepsilon}{2-3\varepsilon}, \qquad \varepsilon_2 = .052, \qquad \varepsilon_{j+1} = 6\varepsilon_j^2, \tag{B1}$$

and

$$0 < r_j \le 1, \quad r_j^{M_j} \ge (1-\varepsilon_j)(1-\varepsilon_{j+1}), \quad U_P \Pi_j = \Pi_j. \tag{B2}$$

These follow from the first row of Eq. (C1) by the doubling estimate of the main text, with a larger starting defect: if $P_{\mathrm{s}}(n, \beta), P_\tau(2n, \beta) \ge 1 - \varepsilon$, move 1 of Eq. (16) gives $P_{\mathrm{s}}(n, 2\beta) \ge (1-\varepsilon)^3$, and squaring with Eq. (12) bounds $1 - P_{\mathrm{s}}(2n, 2\beta)$ by

$$f(1-(1-\varepsilon)^3) = \frac{\varepsilon^2}{2}\left[\sum_{s=1}^{3} (1-\varepsilon)^{-s}\right]^2 \le 6\varepsilon^2 \tag{B3}$$

for $0 \le \varepsilon \le .052$, in place of $5\varepsilon^2$. Since $6\varepsilon^2 \le \varepsilon$, moves 3 and 4 give the same bound for $1 - P_\tau(4n, 2\beta)$. Induction therefore proves $P_{\mathrm{s}}(n_j, \beta_j), P_\tau(M_j, \beta_j) \ge 1 - \varepsilon_j$.

Consider a probability list with purity at least $1 - \varepsilon > 1/2$ and largest weight p_0; put $z = 1 - p_0$. The purity is at most $p_0^2 + z^2 = 1 - 2z(1-z)$, so $2z(1-z) \le \varepsilon$. It is also at most p_0, so $p_0 > 1/2$: the largest weight is unique and $z < 1/2$. Then $z \le 2z(1-z) \le \varepsilon$, hence $z \le \varepsilon/[2(1-\varepsilon)]$ and $z/(1-z) \le d(\varepsilon)$. Applied to the eigenvalues $p_i = |\lambda_i(\beta_j)|^{n_j} / Z_{n_j}(\beta_j)$ of $\sigma_{n_j}(\beta_j)$, the largest-modulus eigenvalue is simple and

$$\sum_{i \ge 1} \left| \frac{\lambda_i(\beta_j)}{\lambda_j} \right|^{n_j} \le d(\varepsilon_j), \tag{B4}$$

which implies Eq. (B1). Positivity of $Z_N(\beta_j)$ for all large odd N ensures that the leading eigenvalue is positive, since a negative leading eigenvalue would make $\mathrm{Tr_v}\, X_{\beta_j}^N$ negative. The residual sum tends to zero since X_{β_j} is Hilbert–Schmidt and its residual spectral radius is strictly smaller than λ_j.

The real kernel admits real orthogonal virtual actions of the three π rotations and the cyclic permutation of the coordinate axes, all commuting with X_{β_j}. On its real one-dimensional leading eigenspace these actions are signs. The order-three cyclic rotation has sign $+1$ and conjugates the three π rotations into one another, so their signs agree. The product of two is the third; their common sign is therefore $+1$. This proves $U_P \Pi_j = \Pi_j$.

Finally $Z_N(\beta_{j+1}) \le Z_N(\beta_j)^2$ by physical purity. Taking Nth roots along even $N \to \infty$ gives $\lambda_{j+1} \le \lambda_j^2$. For the lower bound at the required finite length, the largest eigenvalue of $\sigma_{M_j}(\beta_{j+1})$ is at least $1 - \varepsilon_{j+1}$, since $M_j = n_{j+1}$. Hence

$$\begin{aligned} \lambda_{j+1}^{M_j} &\ge (1-\varepsilon_{j+1}) Z_{M_j}(\beta_{j+1}) \\ &\ge (1-\varepsilon_{j+1})(1-\varepsilon_j) Z_{M_j}(\beta_j)^2 \\ &\ge (1-\varepsilon_{j+1})(1-\varepsilon_j) \lambda_j^{2M_j}, \end{aligned}$$

which is the remaining bound in Eq. (B2).

Appendix C: Starting estimates for the open chain

The certified initial estimates (C1) combine periodic variational estimates with small-cluster boundary certificates from Ref. [2]:

$$P_{\mathrm{s}}(120, 49) \ge .948, \qquad P_\tau(240, 49) \ge .948, \qquad e_{142}(2) \le .14, \qquad P_\tau^{\mathrm{o}}(240, \beta_2) \ge .79. \tag{C1}$$

Combining Eqs. (A16) and (A17) gives

$$e_m(j+1) \le F(q, e_k(j), P_\tau^{\mathrm{o}}(m, \beta_j)), \qquad F(q, e, t) = \frac{q}{1-q}\left[\frac{1}{(1-e)^2 t} - 1\right]. \tag{C2}$$

This function increases with q, e and decreases with t. Track $e_{k_j}(j) \le E_j$ and $P_\tau^{\mathrm{o}}(M_j, \beta_j) \ge 1 - \eta_j$. With $k_2 =$

142 and $k_{j+1} = M_j$, the updates are

$$q_j = d(\varepsilon_{j+1})^{(M_j - k_j)/M_j}, \tag{C3}$$

$$E_{j+1} \geq F(q_j, E_j, 1 - \eta_j), \tag{C4}$$

$$\eta_{j+1} \geq f\big(1 - (1 - \eta_j)(1 - \varepsilon_j)(1 - \varepsilon_{j+1})(1 - E_{j+1})\big). \tag{C5}$$

Choose E_{j+1}, η_{j+1} in $[0, 1)$. Here q_j follows from Eq. (B1) with $n_{j+1} = M_j$, while Eqs. (A18) and (12) determine the purity update. Thus $\varepsilon_2 = .052$, $E_2 = .14$, and $\eta_2 = .21$ start the iteration. Three applications of Eqs. (C4)–(C5) give the following rational upper bounds:

j	E_j	η_j	q_j (upper bound)
2	.14	.21	.142
3	.12	.155	.0282
4	.016	.026	.0028
5	.001	.001	—

Both defects collapse along the iteration, E_j:  and η_j:  ($j = 2, \ldots, 5$, each relative to $j = 2$). Rational arithmetic confirms these step bounds (e.g., $d(\varepsilon_3)^{49} < .142^{120}$ at $j = 2$, with subsequent steps verified by squaring), with $\varepsilon_5 < .000015$.

Induction on scale verifies Eq. (A21) for all subsequent scales: the base case $j = 5$ is given by the table above. Assuming the bound at scale j with $x = x_j \leq .01$, the relation $k_j = M_j/2$ gives

$$\begin{aligned} q_j &\leq \sqrt{\frac{6x^2}{2 - 18x^2}} \leq 2x, \\ F(2x, x, 1 - x) &\leq \frac{6x^2}{(1 - 2x)(1 - x)^3} < 7x^2. \end{aligned} \tag{C6}$$

Setting $E_{j+1} = 10x^2$ satisfies Eq. (C4). The four-factor product in Eq. (C5) is at least $1 - 2x - 16x^2$, and $f(2x + 16x^2) < 3x^2$, so $\eta_{j+1} = 10x^2$ satisfies Eq. (C5) with $\varepsilon_{j+1} \leq 6x^2$, completing the induction for Eq. (A21).